

The Impact of High Blood Pressure on Heart Failure Events: A Propensity Score Matching Analysis

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Introduction

High blood pressure (hypertension) is a well-known major risk factor for cardiovascular diseases. However, analyzing its true causal effect using observational data often encounters selection bias and confounding variables. To address this issue, Propensity Score Matching (PSM) is applied to mitigate these biases, allowing for a more accurate and reliable estimation of the causal impact.

Objectives

1. To evaluate the isolated causal impact of high blood pressure on the occurrence of heart failure events using observational data.
2. To mitigate selection bias and effectively balance baseline clinical characteristics between patient groups using Propensity Score Matching (PSM).

Methodology

Data Source

This study utilized a retrospective observational dataset comprising 25,000 patient records, obtained from the publicly available "Heart Failure Clinical Records" dataset on Kaggle.

Variable Definitions and Statistical Analysis

In this causal inference framework, the variables are defined as follows:

- Treatment Variable (T): High blood pressure status
- Outcome Variable (Y): Occurrence of a heart failure event
- Covariates (X): Baseline confounders including age, sex, cholesterol, maximum heart rate, ST depression (oldpeak), fasting blood sugar, exercise-induced angina, chest pain types, resting ECG, and ST slope.

To control for confounding biases, the propensity score was calculated using a logistic regression model. We then performed a 1:1 nearest neighbor matching algorithm to balance the baseline covariates between the treatment and control groups. Finally, the Average Treatment Effect on the Treated (ATT) was calculated to determine the causal effect.

Results

Before the matching process, the initial study cohort included a total of 25,000 patients. As shown in Table 1, there were 17,344 patients in the normal blood pressure group (Control) and 7,656 patients in the high blood pressure group (Treatment).

Table 1 Characteristics of Heart Failure Patients

Characteristics	Normal BP	High BP	Total	P
N (%)	17,344 (69.40)	7,656 (30.60)	25,000 (100.00)	
Gender, N (%)				0.805
Male	7,782 (44.87)	3,509 (45.83)	11,291 (45.16)	
Female	9,562 (55.13)	4,147 (54.17)	13,709 (54.84)	
Age (years), Median (IQR)	57 (49 – 66)	58 (49 – 65)	58 (49 – 66)	0.655
Cholesterol (mg/dL), Median (IQR)	240 (203 – 277)	239 (202 – 277)	240 (203 – 277)	0.268
Max Heart Rate (bpm), Median (IQR)	162 (146 – 178)	162 (146 – 178)	162 (146 – 178)	0.421
Oldpeak, Median (IQR)	0.80 (0.30 - 1.60)	0.80 (0.30 - 1.50)	0.80 (0.30 - 1.50)	0.216
Heart Failure, N (%)				<.001
Yes	9,845 (56.76)	4,640 (60.61)	14,485 (57.94)	
No	7,499 (43.24)	3,016 (39.39)	10,515 (42.06)	
Fasting blood sugar (>120 mg/dL), n (%)				0.578
Yes	3,788 (21.84)	1,648 (21.53)	5,436 (21.74)	
No	13,556 (78.16)	6,008 (78.47)	19,564 (78.26)	
Exercise-induced chest pain				0.522
Yes	7,000 (40.36)	3,123 (40.79)	10,123 (40.49)	
No	10,344 (59.64)	4,533 (59.21)	14,877 (59.51)	
Chest Pain				0.189
Asymptomatic	8,683 (50.06)	3,767 (49.20)	12,450 (49.80)	
Atypical Angina	3,228 (18.61)	1,403 (18.33)	4,631 (18.52)	
Non-Anginal Pain	3,816 (22.00)	1,712 (22.36)	5,528 (22.11)	
Typical Angina	1,617 (9.32)	774 (10.11)	2,391 (9.56)	
Resting ECG				0.049
LVH	3,458 (19.94)	1,597 (20.86)	5,055 (20.22)	
Normal	10,439 (60.19)	4,482 (58.54)	14,921 (59.68)	
ST	3,447 (19.87)	1,577 (20.60)	5,024 (20.10)	
ST Slope				0.442
Down	1,673 (9.65)	753 (9.84)	2,426 (9.70)	
Flat	7,804 (45.00)	3,497 (45.68)	11,301 (45.20)	
Up	7,867 (45.36)	3,406 (44.49)	11,273 (45.09)	

Abbreviations: ST, S-wave and T-wave; BP – Blood Pressure; Exercise-induced chest pain - ex_angina;

Resting ECG - ecg_cat; ST Slope - slope_cat

To ensure that the PSM process successfully mitigated selection bias, the balance of covariates was evaluated. Table 2 and Figure 1 demonstrate the standardized percentage bias across clinical covariates before and after matching. Post-matching results indicate that all clinical covariates achieved a standardized

percentage bias of less than 5% and p-values greater than 0.05, representing an excellent balance between the two groups.

Table 2. Covariate Balance Test Before and After Matching

Variable	Unmatched (U) Matched (M)	Mean		%bias	P
		Treated	Control		
Sex	U	0.542	0.551	-1.9	0.158
	M	0.542	0.545	-0.7	0.685
Age	U	57.579	57.512	0.6	0.684
	M	57.579	57.437	1.2	0.457
Cholesterol	U	239.110	240.160	-1.9	0.161
	M	239.110	239.000	0.2	0.906
Max heart rate	U	161.780	161.540	1.1	0.433
	M	161.780	161.830	-0.2	0.904
Oldpeak	U	1.093	1.117	-2.3	0.096
	M	1.093	1.099	-0.5	0.746
Fasting blood sugar	U	0.215	0.218	-0.8	0.578
	M	0.215	0.209	1.5	0.343
Exercise-induced chest pain	U	0.407	0.404	0.9	0.522
	M	0.407	0.400	1.6	0.315
Chest Pain	U	1.933	1.905	2.7	0.052
	M	1.933	1.945	-1.1	0.515
Resting ECG	U	1.997	1.999	-0.3	0.820
	M	1.997	1.998	-0.2	0.901
ST Slope	U	2.346	2.357	-1.6	0.235
	M	2.346	2.359	-2.0	0.223

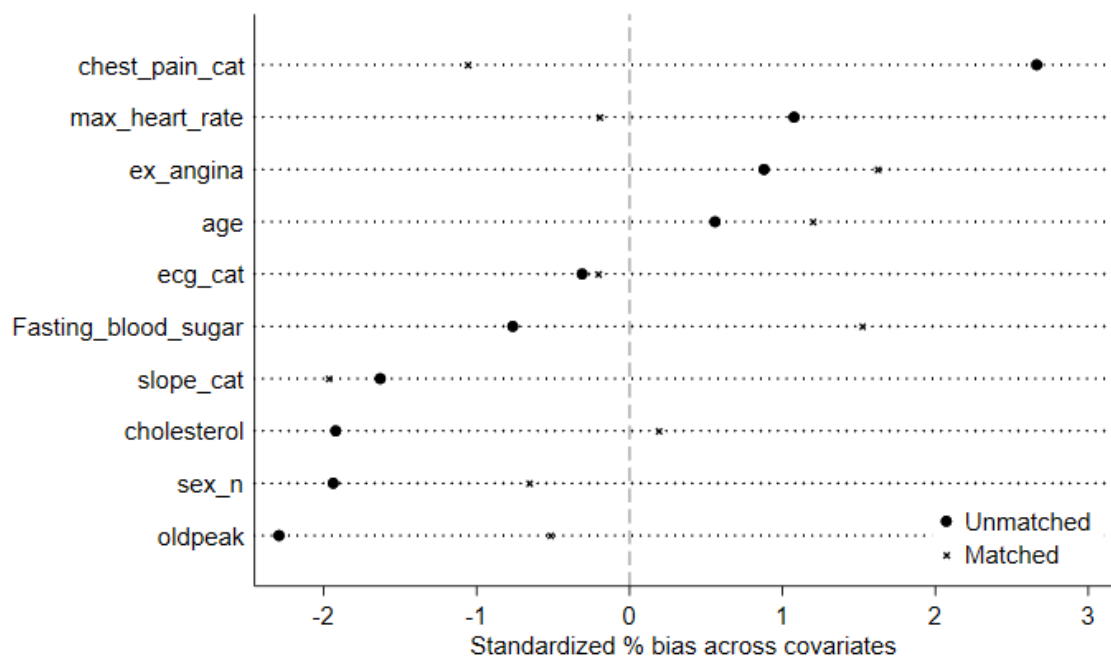


Figure 1 Standardized percentage bias across clinical covariates before (Unmatched) and after (Matched) propensity score matching.

Furthermore, the propensity score distribution is illustrated in **Figure 2**. The complete overlap of the density curves between the treated and control groups confirms an optimal matching quality and an adequate common support region.

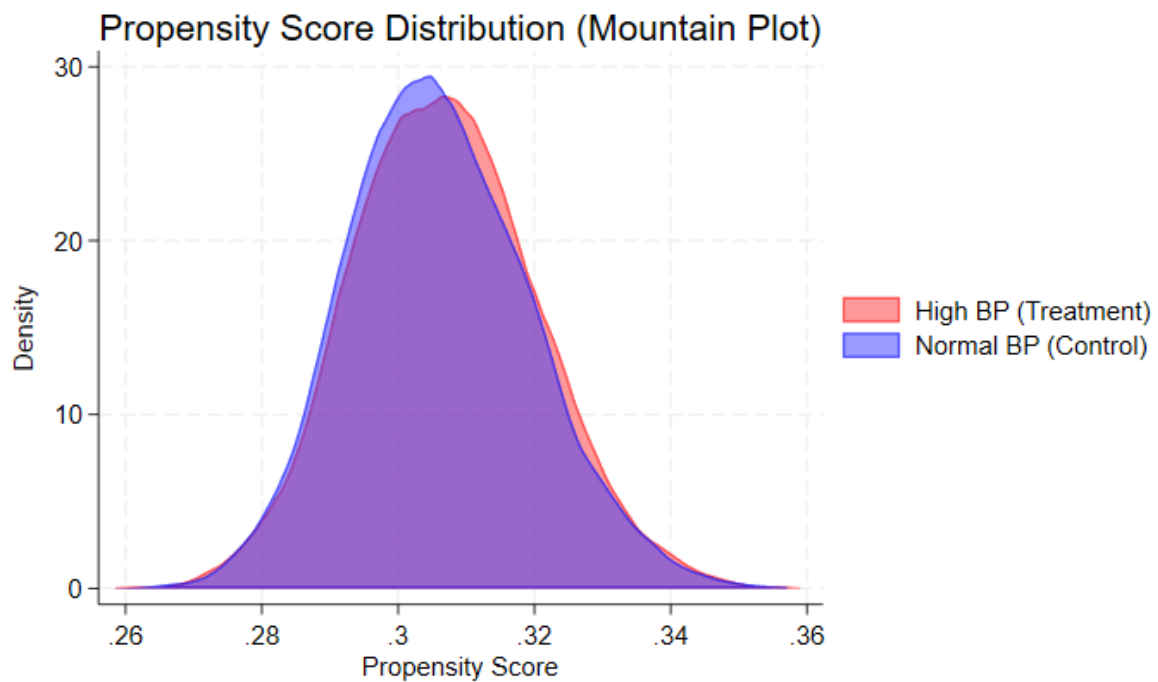


Figure 2. Propensity score distribution and common support region between the high blood pressure (Treatment) and normal blood pressure (Control) groups.

Following the successful matching process, the causal impact of hypertension on heart failure events was estimated. As detailed in Table 3, the analysis revealed that high blood pressure significantly increased the risk of experiencing a heart failure event by 4.21%. This finding is supported by an average treatment effect on the treated of 0.0421, alongside a standard error of 0.0092 and a t-statistic of 4.58.

Table 3 Estimated Effect of Hypertension on Heart Failure Event (ATT)

Variable	Sample	Treated	Controls	Difference	S.E.	t
Heart Failure	Unmatched	0.6061	0.5676	0.0384	0.0068	5.68
	ATT	0.6061	0.5640	0.0421	0.0092	4.58

Conclusions

This study clearly shows that high blood pressure increases the risk of heart failure. We used propensity score matching to balance the patient data and remove bias, which made our comparison fair and reliable. The results reveal that patients with high blood pressure have a 4.21% higher chance of experiencing a heart failure event compared to patients with normal blood pressure. This finding is statistically significant, supported by an average treatment effect on the treated of 0.0421, a standard error of 0.0092, and a t-statistic of 4.58. In short, this analysis proves that high blood pressure directly leads to heart failure.

References

- Lakhani, H. (n.d.). *Heart failure clinical records (25k patients)* [Data set]. Kaggle.
<https://www.kaggle.com/datasets/harshlakhani2005/heart-failure-clinical-records-25k-patients>
- Leuven, E., & Sianesi, B. (2003). *PSMATCH2: Stata module to perform full Mahalanobis and propensity score matching, common support graphing, and covariate imbalance testing*.
<http://ideas.repec.org/c/boc/bocode/s426801.html>